GRADE 3 · DEBUGGING

Trace the Bug

Ontario Math C3.2 — read, predict, and fix code

Name: .... Date: ....

★ **I can trace a loop step by step to find the bug.**

Walk through the code

To find a bug, TRACE the code: go through the loop one repeat at a time and keep track of what Bit has done.

Trace the loop below. After all the repeats, did Bit do what it should? If not, you have found the bug.

Trace this — it SHOULD draw a square

PYTHON

```
for i in range(3):
    bit.forward(50)
    bit.right(90)
```

OUTPUT

```
Trace: side 1, side 2, side 3 … then it stops.
```

1 Trace it

Go through the loop. After repeat 1: 1 side. After repeat 2: 2 sides. After repeat 3: ______ sides. A square needs 4 sides — what is the bug, and the fix?

____

____

2 Trace the counts

A loop is repeat 4 { jump, jump }. Trace the jumps: after each repeat add 2. How many jumps in total? If Bit should only jump 4 times, what is the fix?

____

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3 Predict, then run

Trace for i in range(5): bit.forward(50); bit.right(90). How many sides does Bit draw? Is that right for a square? What is the fix?

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4 Challenge

Trace for i in range(4): bit.forward(50); bit.right(60). It has 4 sides but the shape does not close. The range is right — so what is the bug?

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Big idea

Tracing means walking through the code one repeat at a time. If the end result is not the goal, the place where it goes wrong is the bug.